Name: _____

WORKSHEET

Parallel and perpendicular vectors

1 Show that the following pairs of vectors are perpendicular.

a $(4, -6)$ and $(3, 2)$

b $(8, 6)$ and $(-3, 4)$

c $(4, -3)$ and $(-9, -12)$

d $(15, 25)$ and $(5, -3)$

e $-8\mathbf{i} + 16\mathbf{j}$ and $-6\mathbf{i} - 3\mathbf{j}$

f $(10, -8)$ and $(4, 5)$

g $-16\mathbf{i} - 8\mathbf{j}$ and $2\mathbf{i} - 4\mathbf{j}$

h $-18\mathbf{i} - 9\mathbf{j}$ and $-6\mathbf{i} + 12\mathbf{j}$

i $-\mathbf{i} + 2\mathbf{j}$ and $8\mathbf{i} + 4\mathbf{j}$

© Cengage Learning Australia Pty Ltd 2013

2 Show that the following pairs of vectors are parallel.

a $(4, 10)$ and $(2, 5)$

b $(-12, 24)$ and $(3, -6)$

c $3\mathbf{i} + 2\mathbf{j}$ and $15\mathbf{i} + 10\mathbf{j}$

d $-4\mathbf{i} - 7\mathbf{j}$ and $12\mathbf{i} + 21\mathbf{j}$

e $(1, -4)$ and $(-9, 36)$

f $-16\mathbf{i} - 40\mathbf{j}$ and $-2\mathbf{i} - 5\mathbf{j}$

g $(42, 14)$ and $(6, 2)$

h $-8\mathbf{i} + 16\mathbf{j}$ and $2\mathbf{i} - 4\mathbf{j}$

i $15\mathbf{i} - 27\mathbf{j}$ and $5\mathbf{i} - 9\mathbf{j}$

© Cengage Learning Australia Pty Ltd 2013

3 Determine if the following pairs of vectors are perpendicular, parallel or neither.

a $(-3, 4)$ and $(-21, 28)$

b $(6, 3)$ and $(2, -4)$

c $(-30, 5)$ and $(-5, 30)$

d $(-8, 32)$ and $(-12, -3)$

e $(15, 2)$ and $(-8, 60)$

f $(6, 2)$ and $(7, 14)$

g $(48, 108)$ and $(4, 9)$

h $(-18, -4)$ and $(-3, -24)$

i $(24, 42)$ and $(-8, -14)$

© Cengage Learning Australia Pty Ltd 2013

Answers

1 a $\mathbf{a} \cdot \mathbf{b} = 4 \times 3 + -6 \times 2 = 0$

b $\mathbf{a} \cdot \mathbf{b} = 8 \times -3 + 6 \times 4 = 0$

c $\mathbf{a} \cdot \mathbf{b} = 4 \times -9 + -3 \times -12 = 0$

d $\mathbf{a} \cdot \mathbf{b} = 15 \times 5 + 25 \times -3 = 0$

e $\mathbf{a} \cdot \mathbf{b} = -8 \times -6 + 16 \times -3 = 0$

f $\mathbf{a} \cdot \mathbf{b} = 10 \times 4 + -8 \times 5 = 0$

g $\mathbf{a} \cdot \mathbf{b} = -16 \times 2 + -8 \times -4 = 0$

h $\mathbf{a} \cdot \mathbf{b} = -18 \times -6 + -9 \times 12 = 0$

i $\mathbf{a} \cdot \mathbf{b} = -1 \times 8 + 2 \times 4 = 0$

2 a $\mathbf{a} \cdot \mathbf{b} = 4 \times 2 + 10 \times 5 = 58$; $|\mathbf{a}||\mathbf{b}| =$
$\sqrt{4^2 + 10^2} \times \sqrt{2^2 + 5^2} = \sqrt{116} \times \sqrt{29} = \sqrt{3364} = 58, \therefore \mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|$

b $\mathbf{a} \cdot \mathbf{b} = -12 \times 3 + 24 \times -6 = -180$; $|\mathbf{a}||\mathbf{b}| =$
$\sqrt{(-12)^2 + 24^2} \times \sqrt{3^2 + (-6)^2} = \sqrt{720} \times \sqrt{45} = \sqrt{32400} = 180, \therefore \mathbf{a} \cdot \mathbf{b} = -|\mathbf{a}||\mathbf{b}|$

c $\mathbf{a} \cdot \mathbf{b} = 3 \times 15 + 2 \times 10 = 65$; $|\mathbf{a}||\mathbf{b}| =$
$\sqrt{3^2 + 2^2} \times \sqrt{15^2 + 10^2} = \sqrt{13} \times \sqrt{325} = \sqrt{4225} = 65, \therefore \mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|$

d $\mathbf{a} \cdot \mathbf{b} = -4 \times 12 + -7 \times 21 = -195$; $|\mathbf{a}||\mathbf{b}| =$
$\sqrt{(-4)^2 + (-7)^2} \times \sqrt{12^2 + 21^2} = \sqrt{65} \times \sqrt{585} = \sqrt{38025} = 195, \therefore \mathbf{a} \cdot \mathbf{b} = -|\mathbf{a}||\mathbf{b}|$

e $\mathbf{a} \cdot \mathbf{b} = 1 \times -9 + -4 \times 36 = -153$; $|\mathbf{a}||\mathbf{b}| =$
$\sqrt{1^2 + (-4)^2} \times \sqrt{(-9)^2 + 36^2} = \sqrt{17} \times \sqrt{1377} = \sqrt{23409} = 153, \therefore \mathbf{a} \cdot \mathbf{b} = -|\mathbf{a}||\mathbf{b}|$

f $\mathbf{a} \cdot \mathbf{b} = -16 \times -2 + -40 \times -5 = 232$; $|\mathbf{a}||\mathbf{b}| =$
$\sqrt{(-16)^2 + (-40)^2} \times \sqrt{(-2)^2 + (-5)^2} = \sqrt{1856} \times \sqrt{29} = \sqrt{53824} = 232, \therefore \mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|$

g $\mathbf{a} \cdot \mathbf{b} = 42 \times 6 + 14 \times 2 = 280$; $|\mathbf{a}||\mathbf{b}| =$
$\sqrt{42^2 + 14^2} \times \sqrt{6^2 + 2^2} = \sqrt{1960} \times \sqrt{40} = \sqrt{78400} = 280, \therefore \mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|$

h $\mathbf{a} \cdot \mathbf{b} = -8 \times 2 + 16 \times -4 = -80$; $|\mathbf{a}||\mathbf{b}| =$
$\sqrt{(-8)^2 + 16^2} \times \sqrt{2^2 + (-4)^2} = \sqrt{320} \times \sqrt{20} = \sqrt{6400} = 80, \therefore \mathbf{a} \cdot \mathbf{b} = -|\mathbf{a}||\mathbf{b}|$

i $\mathbf{a} \cdot \mathbf{b} = 15 \times 5 + -27 \times -9 = 318$; $|\mathbf{a}||\mathbf{b}| =$
$\sqrt{15^2 + (-27)^2} \times \sqrt{5^2 + (-9)^2} = \sqrt{954} \times \sqrt{106} = \sqrt{101124} = 318, \therefore \mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|$

© Cengage Learning Australia Pty Ltd 2013

3 **a** Parallel

b Perpendicular

c Neither

d Perpendicular

e Perpendicular

f Neither

g Parallel

h Neither

i Parallel

© Cengage Learning Australia Pty Ltd 2013